is related to the difficulty of the task it was trained on, or to other factors, such as the limited diversity of labels (NLI has only three valid labels) or the type of task.

- **Math** (Sun et al., 2024): this dataset includes both unanswerable and answerable math problems. In our study, we focus exclusively on the answerable problems, as our aim is to assess the correctness of the LLM's outputs, which requires a known correct answer (gold standard). This task introduces an additional, previously unexplored skill of arithmetic reasoning. The train-test split consists of approximately 2,000 and 650 samples, respectively.
- **IMDB** (Maas et al., 2011): contains movie reviews used for the task of sentiment classification.
- **Natural Questions With Context** (Kwiatkowski et al., 2019): the Natural Questions (NQ) dataset is designed to evaluate and train automatic question-answering systems. It consists of real, anonymized queries submitted by users to Google, with answers extracted from Wikipedia, as well as the relevant Wikipedia pages which can be given in context. We included this dataset to introduce an additional challenge that requires adherence to context, complementing the HotpotQA with context dataset.

A.4 Baselines: Implementation Details

Aggregated probabilities / logits. Inspired by prior work (Kadavath et al., 2022; Guerreiro et al., 2023), we compute an aggregated score using the log-probabilities or raw probabilities of the generated text tokens $y_1, y_2, \ldots, y_N$ produced by the generative large language model (LLM). For instance, the following formulation is used to compute the Logits-mean baseline on the entire generated answer:

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{P}(y_i|Q, y_1, ..., y_{i-1}) \tag{1}$$

We also explore aggregation strategies that focus solely on the exact answer tokens (PE-Exact). Following Varshney et al. (2023), we also experiment with aggregating the minimum and maximum values (PE-[Min—Max]-[Exact]), alongside the mean aggregation described in Equation 1.

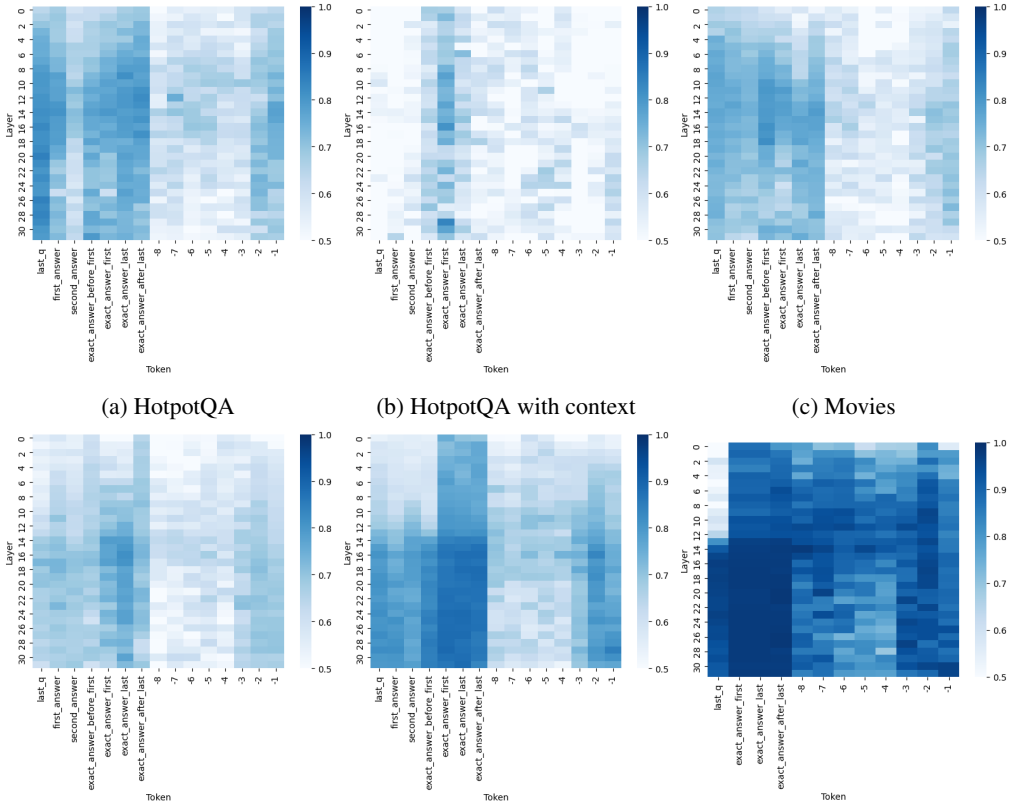
P(True): We follow Kadavath et al. (2022) and prompt the LLM to judge whether its answer is correct. Our prompt followed the following template, from Kadavath et al. (2022):

> Question: `[Question]`
>
> Proposed Answer: `[LLM long answer]`
>
> Is the proposed answer:
>
> (A) True
>
> (B) False
>
> The proposed answer is:

B Full Error Detection Results

Figure 6 presents the AUC values of a traind probe across layers and token for Mistral-7b-instruct, showing a similar pattern across all datasets. We also observe similar patterns across other models. See our repo `https://github.com/technion-cs-nlp/LLMsKnow`for the figures.

Tables 4, 5, 6, and 7 present the full error detection results across all baselines and datasets, which are consistent with the main paper results.



(a) HotpotQA (b) HotpotQA with context (c) Movies

(d) Winogrande (e) NLI (f) IMDB

Figure 6: AUC values of a probe error detector across layers and tokens, Mistral-7b-instruct. The detection performance spikes at the exact answer tokens.

Table 4: Comparison of error detection performance (AUC) on Mistral-7B.

	Mistral-7B				
	TriviaQA	Winobias	Math	Movies	IMDB
Logits-mean	0.67 ±0.004	0.49 ±0.010	0.41 ±0.015	0.67 ±0.007	0.88 ±0.064
Logits-mean-exact	0.67 ±0.004	0.50 ±0.010	0.56 ±0.026	0.68 ±0.008	0.57 ±0.080
Logits-min	0.80 ±0.003	0.45 ±0.014	0.48 ±0.021	0.73 ±0.006	0.78 ±0.056
Logits-min-exact	0.80 ±0.005	0.53 ±0.014	0.78 ±0.032	0.72 ±0.005	0.57 ±0.080
Logits-max	0.53 ±0.008	0.49 ±0.010	0.42 ±0.023	0.54 ±0.005	0.83 ±0.076
Logits-max-exact	0.54 ±0.009	0.50 ±0.010	0.40 ±0.024	0.58 ±0.007	0.57 ±0.080
Probas-mean	0.76 ±0.003	0.53 ±0.018	0.66 ±0.016	0.72 ±0.007	0.87 ±0.041
Probas-mean-exact	0.78 ±0.002	0.55 ±0.014	0.62 ±0.016	0.74 ±0.007	0.83 ±0.057
Probas-min	0.82 ±0.003	0.52 ±0.013	0.82 ±0.020	0.73 ±0.006	0.86 ±0.032
Probas-min-exact	**0.85** ±0.003	0.58 ±0.011	0.84 ±0.015	0.74 ±0.006	0.83 ±0.057
Probas-max	0.53 ±0.008	0.50 ±0.016	0.43 ±0.025	0.55 ±0.008	0.80 ±0.074
Probas-max-exact	0.55 ±0.009	0.51 ±0.013	0.39 ±0.019	0.59 ±0.009	0.83 ±0.057
p(True)	0.57 ±0.007	0.53 ±0.019	0.56 ±0.027	0.51 ±0.003	0.65 ±0.004
p(True)-exact	0.56 ±0.006	0.55 ±0.026	0.57 ±0.036	0.52 ±0.003	0.65 ±0.003
Probe @ token					
Last generated [-1]	0.83 ±0.002	0.65 ±0.008	0.82 ±0.023	0.79 ±0.002	0.85 ±0.007
Before last generated [-2]	0.82 ±0.003	0.84 ±0.012	0.83 ±0.019	0.78 ±0.003	0.95 ±0.004
End of question	0.74 ±0.005	0.78 ±0.012	0.83 ±0.016	0.77 ±0.002	0.81 ±0.009
Exact answer last	0.84 ±0.005	**0.89** ±0.007	**0.96** ±0.008	0.78 ±0.003	**0.95** ±0.004
Exact answer last+1	0.84 ±0.004	0.84 ±0.012	0.95 ±0.010	**0.80** ±0.002	0.85 ±0.007
	HotpotQA	HotpotQA-WC	Winogrande	NLI	NQ-WC
Logits-mean	0.63 ±0.005	0.52 ±0.009	0.49 ±0.004	0.51 ±0.004	0.69 ±0.006
Logits-mean-exact	0.57 ±0.008	0.52 ±0.007	0.50 ±0.003	**0.93** ±0.004	0.72 ±0.005
Logits-min	0.72 ±0.008	0.59 ±0.006	0.50 ±0.007	0.53 ±0.005	0.65 ±0.009
Logits-min-exact	0.72 ±0.007	0.65 ±0.004	0.51 ±0.007	0.49 ±0.006	0.70 ±0.005
Logits-max	0.54 ±0.007	0.49 ±0.010	0.48 ±0.005	0.48 ±0.005	0.59 ±0.012
Logits-max-exact	0.48 ±0.010	0.44 ±0.007	0.50 ±0.003	0.48 ±0.005	0.58 ±0.009
Probas-mean	0.65 ±0.004	0.55 ±0.006	0.51 ±0.007	0.49 ±0.003	0.63 ±0.008
Probas-mean-exact	0.62 ±0.006	0.56 ±0.007	0.51 ±0.005	0.02 ±0.001	0.66 ±0.007
Probas-min	0.73 ±0.005	0.58 ±0.007	0.52 ±0.009	0.53 ±0.004	0.63 ±0.011
Probas-min-exact	0.78 ±0.005	0.66 ±0.004	0.52 ±0.008	0.49 ±0.005	0.69 ±0.006
Probas-max	0.54 ±0.008	0.49 ±0.007	0.50 ±0.005	0.47 ±0.004	0.52 ±0.004
Probas-max-exact	0.48 ±0.010	0.44 ±0.005	0.50 ±0.004	0.48 ±0.003	0.53 ±0.012
p(True)	0.55 ±0.007	0.54 ±0.006	0.51 ±0.005	0.51 ±0.003	0.52 ±0.008
p(True)-exact	0.61 ±0.005	0.54 ±0.006	0.61 ±0.006	0.51 ±0.006	0.53 ±0.014
Probe @ token					
Last generated [-1]	0.78 ±0.006	0.67 ±0.004	0.51 ±0.007	0.77 ±0.004	0.78 ±0.003
Before last generated [-2]	0.79 ±0.007	0.69 ±0.007	0.66 ±0.004	0.81 ±0.002	0.75 ±0.006
End of question	0.72 ±0.007	0.56 ±0.003	0.51 ±0.007	0.88 ±0.004	0.70 ±0.005
Exact answer last	0.80 ±0.008	**0.74** ±0.007	**0.69** ±0.006	0.84 ±0.004	0.81 ±0.009
Exact answer last+1	**0.81** ±0.008	0.72 ±0.005	0.59 ±0.005	0.75 ±0.006	**0.84** ±0.007